Clinical Risk Assessment Report: Patient 2

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 14.6% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.66x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Minimum Tumor Density (CONVENTIONAL_HUmin), which reduced the patient's absolute risk by 1.7%. Biologically, this feature reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: 0.27 [Avg (61th %ile)] (+1.6%)**

Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.7 [Avg (28th %ile)] (+1.3%)**

Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

3. Protective / Mitigating Factors (Reducing Risk)

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.66 [Bottom 24%] (-1.7%)**

Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Large Zone Emphasis (CT) (GLZLM_LGZE.CT) **Value: -0.22 [Avg (46th %ile)] (-1.4%)**

Reflects the presence of large homogeneous areas; lower values suggest a chaotic, heterogeneous microenvironment.

Long Run High Gray-Level Emphasis (PET) (GLRLM_LRHGE.PET) **Value: 0.03 [Avg (70th %ile)] (-1.1%)**

Points to continuous, elongated bands of high-density or highly active tumor tissue.

Short Run Low Gray-Level Emphasis (CT) (GLRLM_SRLGE.CT) **Value: -0.14 [Avg (47th %ile)] (-0.6%)**

Indicates fragmented areas of low density, often associated with diffuse micro-necrosis or loose tissue structure.

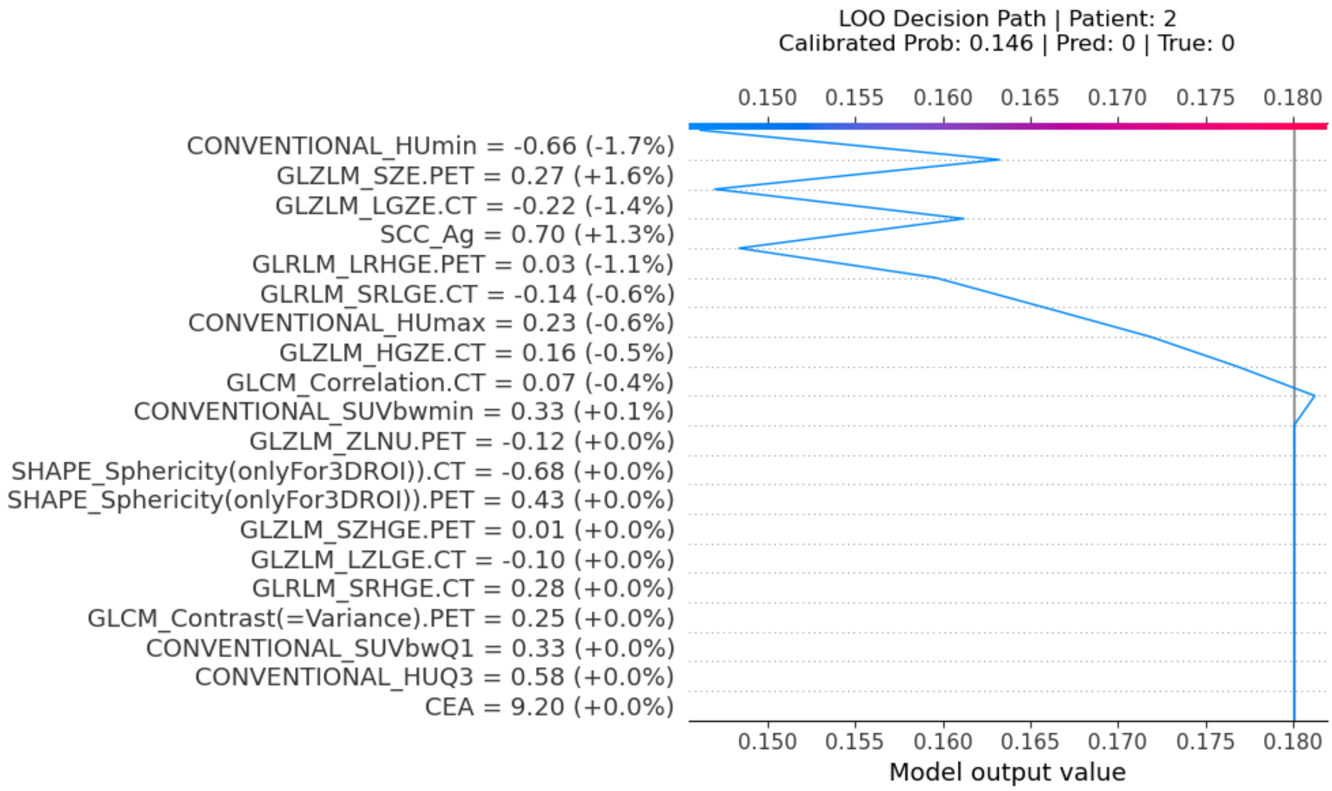
Maximum Tumor Density (CONVENTIONAL_HUmax) **Value: 0.23 [Avg (60th %ile)] (-0.6%)**

Reflects the most dense solid components or calcifications within the primary tumor lesion.

High Gray-Level Zone Emphasis (CT) (GLZLM_HGZE.CT) **Value: 0.16 [Avg (49th %ile)] (-0.5%)**

Highlights the distribution of high-density or highly metabolic active regions clustered within the tumor.

Machine Learning Feature Attribution (SHAP Decision Path)



Sensitivity Analysis: Feature Tipping Points

The plots below demonstrate how variations in specific clinical or radiomic features affect the model's predicted risk. The red marker indicates the patient's current clinical state. Crossing the red dashed line changes the diagnosis.